Advanced Animation in After Effects

In the last lab we learned about the basics of creating an After Effects project including importing files, creating compositions and animating layer properties. In this lab we will continue to learn animation techniques with Adobe After Effects. In this lab we will animating a spooky Halloween scene.

Overview

Take a moment to look at the video will will create using Adobe After Effects. You can view the video on YouTube below:

https://www.youtube.com/watch?v=iYdQTFyXj24

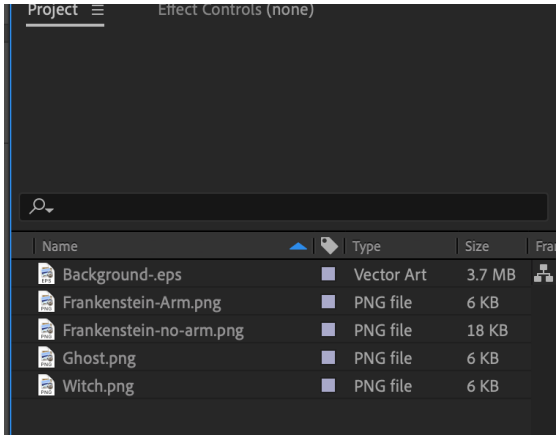
Make sure you have downloaded the lab files needed to complete this lab. The files are in zip folder called *After-Effects-Advanced.zip*. Make sure to unzip the folder completely to use the files inside appropiately.

Create New Project / Import Files

Open Adobe After Effects and create a new project from the start screen.

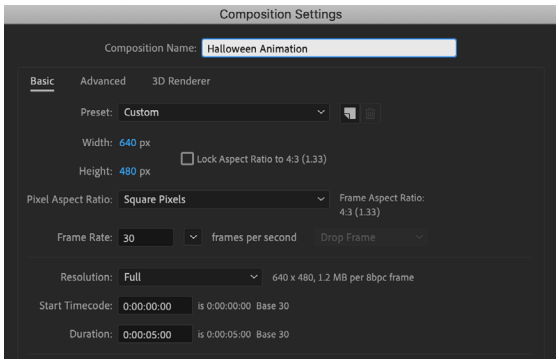
In the *Project* tab import the following files into your project.

- Background.eps
- Frankenstein-Arm.png
- Frankenstein-No-Arm.png
- Ghost
- Witch



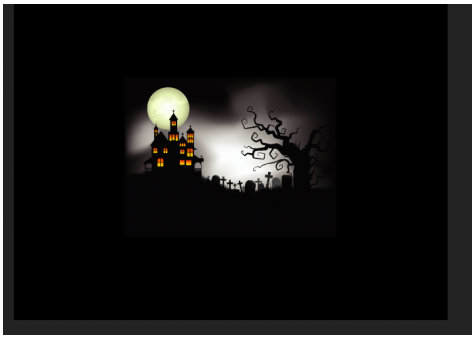
Create a new composition called *Halloween Animation* with the following settings:

```
Width: 640px
Height: 480px
Frame Rate: 30
Pixel Aspect Ratio: Square
Duration: 00:00:05:00
```



Layout the Scene

Now that we have our files and composition ready to go, let's begin by building our scene. Drag *Background.eps* into the *Compositionn* window. We want the background image to fill the entire background so let's scale it to fill the entire *Composition* screen.



After scaling the *Background.eps* layer you may notice that the

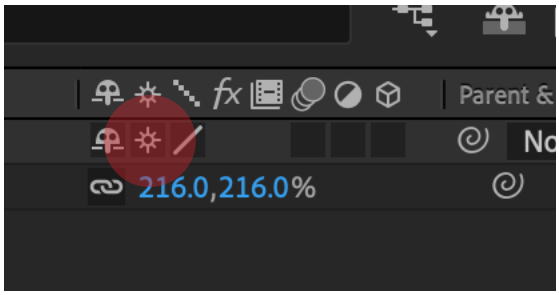
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quality of the image looks very poor. Like Adobe Photoshop, video and motion graphics programs are pixel / raster graphic programs. In a raster program when a graphic is scaled up it begins to distort and become blury and pixelated. Because the *Background.eps* file was scaled up it has began to distort and become blurry.



There is a possible solution to this issue. In addition to raster graphics there is something called vector graphics. Vectors graphics unlike raster graphics are not built with pixels, but instead are built using mathematical equations. Vector graphics can be built in vector graphic programs like Adobe Illustrator, and Figma. One benefit of vector graphics compared to raster graphics is that vector graphic can be scaled without distortion. If a graphic is saved in a vectorm file format as such as *.ai*, .or*.eps* than it will keep this ability.

Our *background.eps* is an EPS or vector format. *"But why does it look distorted, I thought you said vector graphics don't become distorted when scaled?"* you may be thinking. The reason is by default vector graphics in Adobe After Effects act as raster graphics. That being said, there is a way to tap into the vector graphic power, to do this in the *Timeline* click on the checkbox marked *Continously Rasterize*, it looks like as asterik mark. After clicking the checkbox you should notice that the background imageappears much clearer, that because this option allows the *Background.eps* to act like a vector, instead of as a raster graphic, and this being scaled up or down will not cause distortion. You can see how building a project with vectors graphics can be advantaneous to prevent future quality problems.



Lock the *Background.eps* in the timeline so we don't accidenly move or alter it.

Save your Project as *lastname-advanced-animation.eps.*

Animate the Ghost

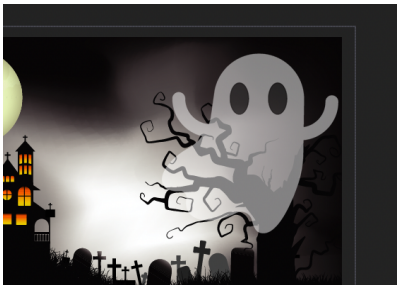
Now that we have the background of our scene let's begin to animate the other elements in the scene. Drag the *Ghost.png* file into the *Composition* window and place it at the top right corner of the scene. Use the image on the right as a reference, don't worry if it is not 100% exact.



The ghost character is fun, but does not look to "ghostly." We can help with this by playing around the *Opacity* property. The *Opacity* property here is similar to how the *Opacity* property works in programs like Adobe Photoshop and Adobe Illustrator in that it controls how transparent or see-through the layer is.

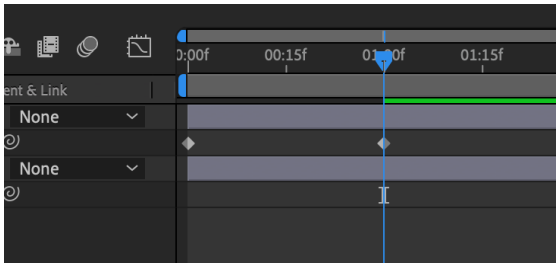
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Select the *Ghost* layer in the timeline and press *"T"* to access the *Opacity* property, the *"T"* in this case is for for transparency. You can see the *Ghost.png* is set to *100%* or 100% opaque. To make the Ghost more see-through lowere the *Transparency* option to *50%*, now that ghost feels a bit more like a ghost.



Let's take it a step further, let's animate the opacity of the ghost so that it starts invisible and then fades in to *50%* after 1 second. To do this we'll animate the *Opacity* property using *Keyframes*.

In the timeline move the time marker to the beginning at *00:00:00:00*. Select the *Ghost.png* layer and set the transparency property to *0%*. Click the stopwatch to begin auto keyframing the property. Remember now that the keyframing has been enabled any time you change the this property in the timeline it will automatically make a saved point keyframe.

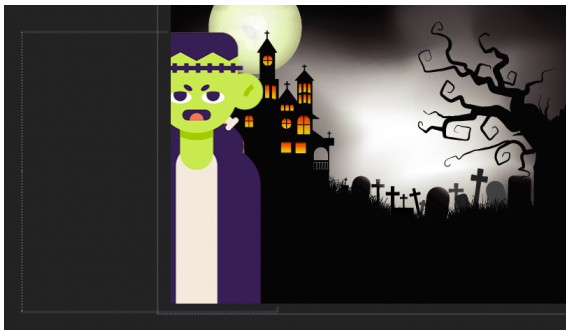


Take the time marker and move it to one second into our animation or *00:00:01:00* in the timeline. Change the *opacity* property to *50%*. Play the animation from the start and you should see the Ghost character start invisible and then fade in after one second of animation.

So what exactly happened here and how did the keyframes help? Remember that the keyframes represent saved points in time. The keyframe at frame 1 saved the Ghost's opacity at *0%*. The second keyframe at one second of the animation saved the Ghost's opacity at 50%. In otherwords, After Effects was told that the Ghost's opacity had to change from *0%* to *50%* in one second. The result is the ghost fades in over one second.

Save your project.

Animate Frankenstein



Now that the ghost is animating let's move on to Frankenstein. In this scene we are going to make Frankenstein wave his arm.

Begin by dragging *Frankenstein-No-Arm.png* into the *Composition* window. Use the *Selection Tool* to place Frankenstein to be half on and half off the left hand side of the *Composition* window. Use the image on the right as a reference, don't worry about being 100% exact. Use the *timeline* lock function to lock this layer into place.



Next drag *Frankenstein-Arm.png* into the *Composition* window. Place it so that the arm matches up with Frankenstein's shoulder. Make sure that the *Frankenstein-Arm.png* layer is **above** the *Frankenstein-No-Arm.png* layer in the layer order.

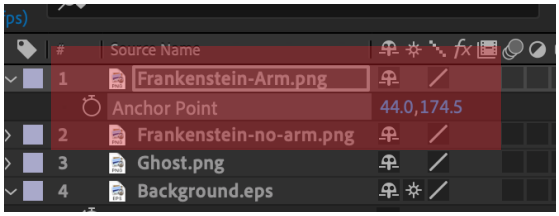
To make Frankenstein wave his arm let's use the rotation property we learned about in the previous lab. Select the Frankenstein-

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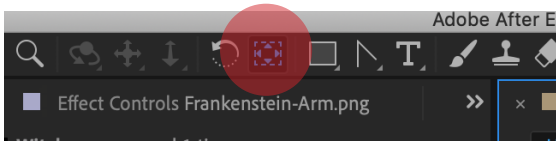
Arm layer and hit the *'R'* key on the keyboard to show the rotation property. Begin to use the degree and you'll notice that the arm is acting a bit odd. The arm is rotating, but rather than rotating from the shoulder it is rotating from the center of the arm. The reason for this has to do with something called the *anchor point*.



The *anchor point*, sometimes referred to as a pivot point, is the point at which an object rotates. Sometimes that pivot point is in the center of the object, like rotating a wooden block, sometimes it is not. For example think of the Earth. The Earth revolves around the sun, thus, the pivot point of the Earth would be the sun, not the center of the Earth itself. In the case of Frankenstein's arm, the pivot point would be the shoulder, not the center of the arm. In order to change the pivot point or anchor point we will change the anchor point for the arm.



Make sure that *Frankenstein-Arm.png* is selected in the timleline and hit the *'A"* key on the keyboard to open the *anchor point* property. The anchor point property has two values, horizontal and vertical. These two points represent the pivot point of the layer.



Zoom in on *Frankenstein-Arm.png* layer and you should see an icon that looks like a cross-hair, that is the *anchor point*. We want to move that cross-hair icon to top of the arm, where the arm would attach to the shoulder. This can be done either by adjustment the anchor point property values or you can select the *Pan Behind* option in the toolbar. After selecting the *Pan Behind* tool click and drag the cross-hair to adjust the anchor point.



Now that the anchor point has be readjusted the arm may be slightly out of place. Use the *Selection Tool* to click and drag the arm to connect with the shoulder of Frankenstein like the image on the right. Use the *rotation* property to rotate the arm and you should see the arm rotating from the shoulder as opposed to the center.

Multiple Keyframes

So far in our animation we've only animated using two keyframes, start-to-finish. However, in a complicated animation you are likely to use multiple keyframes to animate a layer.

Now that Frankenstein's arm is in place let's begin to have him wave. The wave will essentially be him raising his arm, lowering his arm and then raising and lowering it again. To do this we'll need use multiple keyframes.

Use the time marker and place it at the beginning of the timeline. Select the *Frankenstein's Arm* layer and use the rotation property. Set the arm to be down. Click the stopwatch on the rotation

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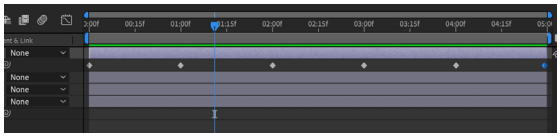
property to begin auto keyframing.

Move the time marker to the 1 second position in the timeline or *00:00:01:00*. Use the *rotation* property to adjust the arm to move upward and create a second keyframe. Play back the animation and you should see Frankenstein's arm rotate upward. That's a good start, but as you can see in the sample video Frankenstein moves his arm up and down 2.5 times.

Move the time marker to the 2 second position in the timeline or *00:00:02:00*. Use the *rotation* property to adjust the arm back downward, close to where it began at the beginning of the animation to create a third keyframe. Play back the animation and you should see Frankenstein's arm rotate upward and then rotate downward.

So what exactly happened here? Like in the past After Effects took the saved points of keyframe 1 and 2. It rotated Frankenstein's arm from the *rotation* value of keyframe 1 and ended at the *rotation* value of keyframe 2. Then what happened is the animation continued. Keyframe 2 became the beginning *rotation* value of the next animation from keyframe 2 to keyframe 3. The more keyframes you layer has the more actions will occur.

Use the *rotation* property to adjust Frankenstein's arm on the 3 second position, 4 second position and 5 second position. The animation should have Frankenstein's arm rotating up and down. Use the video example as a reference. If you make a mistake with your keyframe remember you can always edit the keyframe value or highlight the keyframe using the *Selection Tool* and delete the keyframe using the *delete / backspace* key on the keyboard. Don't worry is the animation is not 100% exact to the video example.



Save the animation

Curved Animation Paths

The last topic we will discuss today is the idea of curved animation paths. In the past when we have animated the position of a layer the movement went in a straight linear line, we will learn how to adjust the paths to make them curved.



Begin by dragging *witch.png* into the *Composition* window. Use the *Selection Tool* to place the witch at the lower right hand corner of the scene as shown in the image on the right, don't worry if your is not 100% exact to mine.

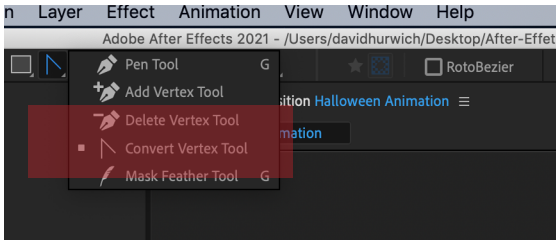
Keyframe the *position* property to have the witch go from the lower right hand side of the *Composition* window to the upper left hand. The animation should start at the beginning of the timeline and end at the end of the timeline.

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When you play back the animation the witch will move in a linear path from the lower right to upper left hand side of the composition. However, we want a bit of pizazz, so let's curve her motion path.



To edit the motion path of the witch we can going to use something called a *Bezier handle*. Bezier handles are handles that allow users the ability to edit and modify a vector curve, in this case the motion path of the witch is the vector curve. To access the Bezier handles we are going to use the *Convert Vertex Tool*. The *Convert Vertex Tool* is located under the *Pen Too*l in the toolbar.



Using the *Convert Vertex Tool* click on the squares icon of the witch motion path. The square represents the keyframe in Adobe After Effects, but it is also something we call in vector graphic an anchor (not to be confused with the anchor point). Click and drag using the *Convert Vertex Tool* on the two squares on the motion path to create Bezier handles. Use the Bezier handles to adjust the motion line to make it a curve angled upward. Use the image on the right as a reference, don't worry if your is not 100% exact to the image. Play the animation and you should see the witch curve upward instead of going in straight linear path. Adjust the Bezier handles to adjust the curve if you want to edit the curve itself.

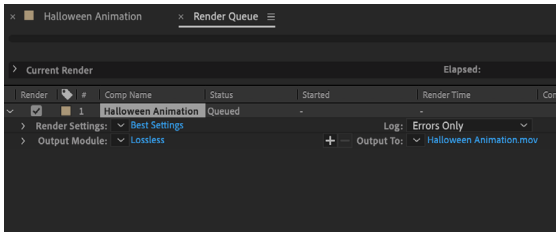


Finally, let's add a bit of perspective. As objects go into the distance they get smaller. As the witch falls into the distance in front of the moon she would get smaller. To achieve this affect use the scale property to scale the witch to scale down. The scal should start at the beginning of the timeline and end at the end of the timeline. Use the *"U"* key to view both the scale and position properties.

Finally, adjust the witch layer so it above the *Background.eps* layer, but below the *Frankenstein* and *Ghost* layers.

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Export



Great job! We covered alot here. With *Halloween Animation* composition open in the timeline go to *Composition > Add to Render Queue*.

In the *Render Queue* look for *Comp Name* make sure it is set to *Halloween Animation*.

Under *Render Settings* make sure it is set to *"Best Settings".*

Under *Output Module* make sure it is set to *"Lossless"* make sure the format is set to either Quicktime or AVI.

Finall, use the *Output To* option save the animation to your Desktop with the label *lastname-Animation-Advanced.*

Submission

Upload your exported animation video format to your YouTube account. Publish the video and you will copy and paste the video URL to the submission page for the lab.

In addition to the final video url please upload the *lastname-advanced-animation.aep* file to Canvas for grading. Please do NOT submit all of the files or videos as those are large files, instead just upload the *.aep* project file for grading purposes.

Extra Credit: Not Required

+2 points

The following is extra credit and is not required as part of the lab.

Import *bat.eps into your project.* Animate at least three separate version of *bat.eps* moving across the composition. Each bat should have a curved motion path.

Export your animation including the bat characters.

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